EE3032 Introduction to VLSI Design—Final

1. (30 points) Answer the following questions.
   (a) List two requirements for drawing the layout of a cell in the cell library. (4 points)
   (b) The dynamic power dissipation (switching power) of a complex gate can be divided two power components. List the two power components. (4 points)
   (c) List two design rules for avoiding glitch power of a CMOS circuit. (4 points)
   (d) List two approaches for coping with the power and ground bounce. (4 points)
   (e) List two major problems of the dynamic logic. (4 points)
   (f) List two limitations of the CMOS domino logic. (4 points)
   (g) What are the core-limited design and the pad-limited design? (4 points)
   (h) What is the physical design? (2 points)
2. (15 points) Figure 1 shows a register realized by two low-cost latches. Complete the following questions.
   (a) What type of the register is? Explain your answer. (5 points)
   (b) Assume that the switching threshold voltage of the inverter Inv1 is $V_{dd}/2$ and the turn-on equivalent resistances of the transistors N1, N2, P1, and P2 are R_{N1}, R_{N2}, R_{P1}, and R_{P2}, respectively. List two inequalities for determining the sizes of P2 and N2. (10 points)

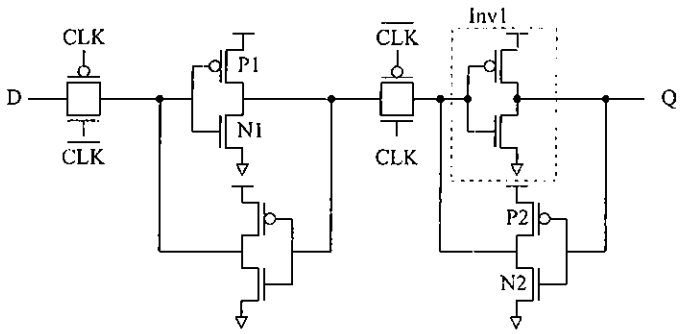


Figure 1: A register consists of two low-cost latches.

3. (15 points) Sketch the Boolean function $f = \overline{AB + DC + CE}$ using each of the following logic design techniques: (1) Dynamic logic (5 points), (2) Fully differential logic (5 points), and (3) Static CMOS logic. (5 points)
4. (15 points) Figure 2 shows a dynamic logic gate. Assume that $\mu_n = 2\mu_p$ and the size W/L of the P1 is $4\mu m/0.18\mu m$. Also, the gate capacitance (C_g) of the P1 is C, i.e., $C_g = C$. Complete the following questions.
   (a) Determine the sizes of the NMOS transistors X1, X2, X3, X4, X5, X6, and N1 such that the rise time and the fall time of the logic gate are equal, i.e., $t_r = t_f$. (5 points)
   (b) Assume that the gate capacitance of a PMOS or an NMOS is proportional to the gate area of the PMOS or the NMOS. If your boss ask you to design the dynamic gate such that the switching power of the clock (CLK) is smaller than $1.5fCV_{dd}^2$, where f and V_{dd} denote the operation frequency and the power supply of the logic gate. Determine the sizes of the NMOS transistors X1, X2, X3, X4, X5, X6, and N1 such that the rise time and the fall time of the logic gate are equal, i.e., $t_r = t_f$. (10 points)

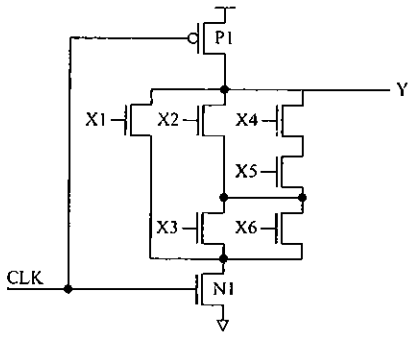


Figure 2: A dynamic logic circuit.

5. (15 points) Figure 3 shows a combinational circuit designed with NP-domino logic. Assume that all the NMOS transistors have the same β_n and all the PMOS transistors have the same β_p, and $\beta_n = \beta_p$. Also, $\mu_n = 2\mu_p$ and the gate capacitance of the PMOS and NMOS transistors are $2C$ and C, respectively; and the turn-on equivalent resistance of the NMOS transistor or the PMOS transistor is R. Complete the following questions.

   (a) Assume that the rise time t_r or fall time t_f of a logic gate can be estimated by the product of the output capacitance of the logic gate and the equivalent resistance of the output charge path or the output discharge path, respectively. Also, the delay t_d of a logic gate can be calculated by $t_d = (t_r + t_f)/2$ and the drain capacitance of the transistors can be neglected. Calculate the contamination delay t_{cd} and the propagation delay t_{pd} of the combinational circuit in terms of R and C. (10 points)

   (b) Assume that the V_{dd} of the combinational circuit is $1.8V$ and the threshold voltage of the PMOS transistor is $|V_{tp}| = 0.6V$. Also, the capacitance of the nodes x, y, and z are 0.4pf, 0.8pf, and 2pf, respectively. Can the combinational circuit be operated correctly? Explain your answer. (5 points)

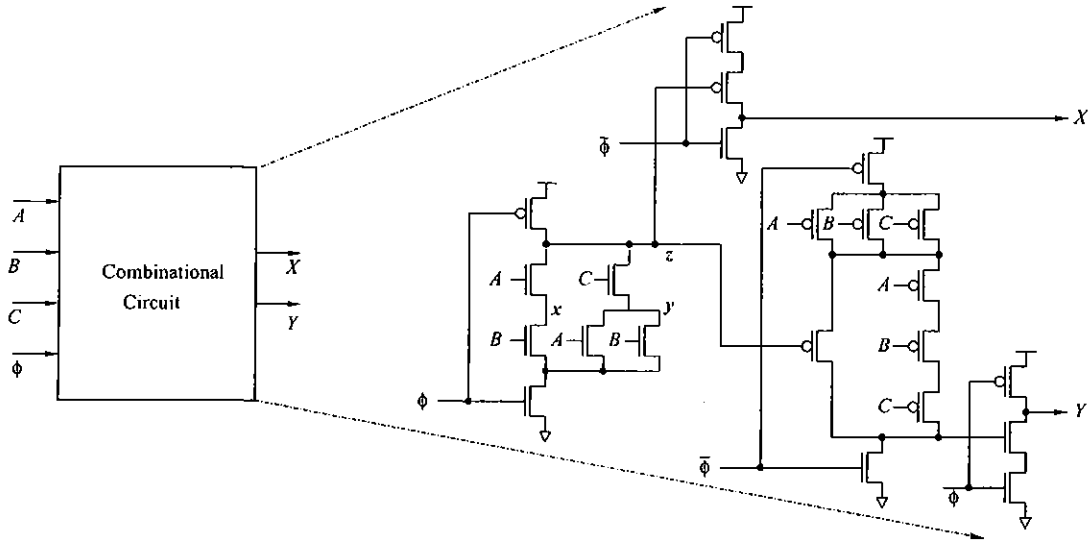


Figure 3: A circuit designed with NP-domino logic.

6. (10 points) Figure 4(a) shows a pipelined circuit. Assume that your boss asks you to design the circuit such that the circuit can be operated at 100MHz. Also, the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers R1, R2, and R3 are $0.8ns$, $0.4ns$, $1.6ns$, and $1.2ns$, respectively. Complete the following questions.

(a) If the clock skew of the CLK is 0.1ns, determine the maximum delay of the combinational circuit. (5 points)

(b) If the delay of the interconnection between the R2 and R3 is 0.2ns and the delay of the buffer shown in Figure 4(b) is 0.3ns, then how do you design the pipeline circuit such that it can be operated correctly? Draw your modified pipelined circuit. (5 points)

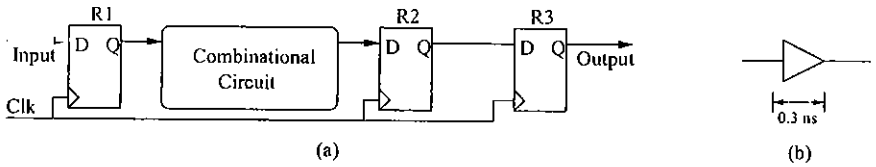


Figure 4: (a) A pipelined circuit. (b) A buffer.

7. (15 points) Figure 5 shows the symbolic layout of a circuit. Draw its transistor-level schematic diagram.

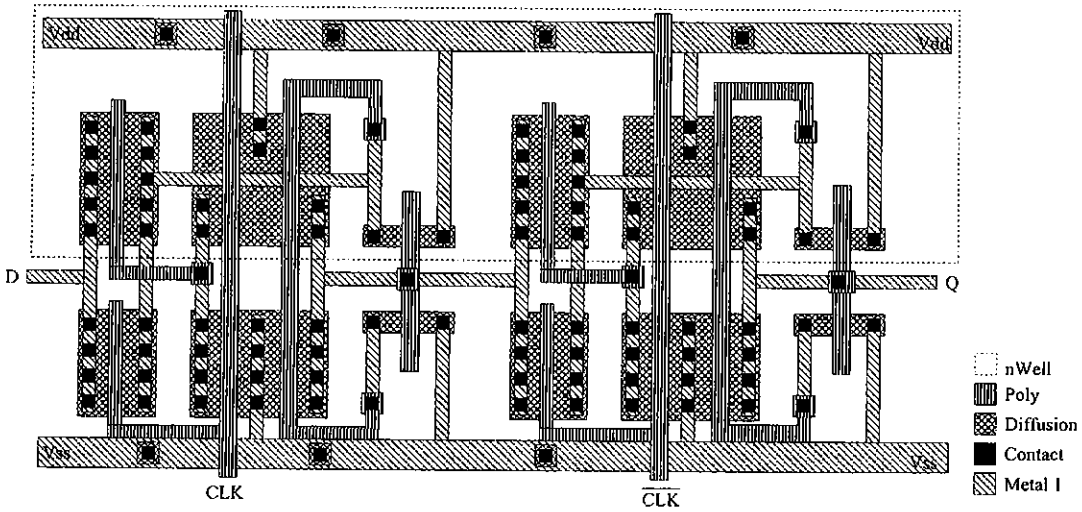


Figure 5: The symbolic layout.

8. (15 points) Figure 6 shows the Schmitt CMOS circuit and its voltage transfer curve. Assume that the supply voltage $V_{dd} = 1.8V$ and the threshold voltage of the NMOS transistors is $V_{tn} = 0.6V$. Your boss ask you to design the Schmitt CMOS circuit such that its $V_{T+} = 1.2V$. Please derive the ratio of the β_{N_1} to the β_{N_3}.

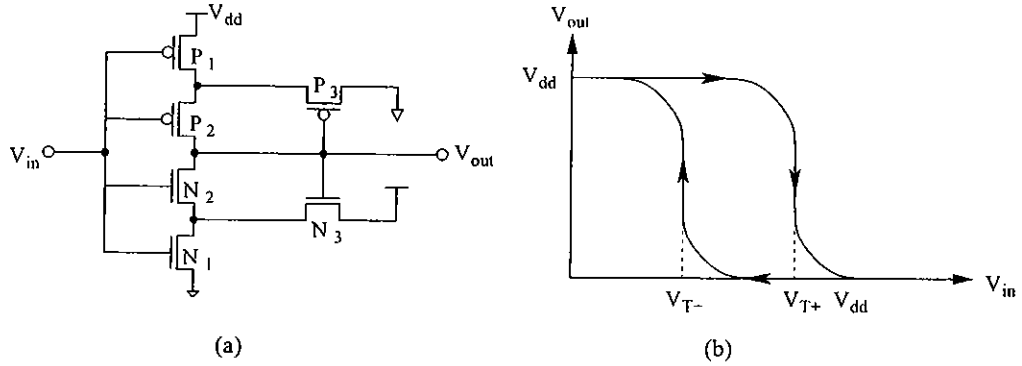


Figure 6: (a) The CMOS Schmitt circuit. (b) The voltage transfer curve the the Schmitt circuit.